QuestionsLab1.2.pdf

abcd__1234

Paralelismo

3º Grado en Ingeniería Informática

Facultad de Informática de Barcelona (Fib)
Universidad Politécnica de Catalunya

Qualificació **3,00** sobre 3,00 (**100**%)

Pregunta 1
Completa
Puntuació 1,00 sobre 1,00
Marca la pregunta

Which of the following affirmations are true about Tareador:

- [ ] Tareador is a tool that is used to find bugs in the specification of potential tasks. To that end, Tareador executes the application in parallel when the programmer hits the “View simulation” button specifying a certain number of processors.
- [x] Tareador sequentially executes the application, properly annotated to identify potential parallel tasks. It includes a simulation component to understand how it would execute in parallel when using a certain number of processors.
- [x] Tareador records all memory allocations and memory accesses performed by annotated tasks and counts the number of instructions that are performed by them in order to build the TDG that characterizes the annotated task decomposition.
- [ ] Tareador visualises the TDG that is generated after the execution of an application annotated using tareador_start_task and tareador_end_task. It shows task granularities (execution time for tasks) and the order in which the tasks have been executed.
- [x] When Paraver is invoked from Tareador to display the timeline with the simulated parallel execution: the horizontal axis represents time assuming that each instruction in the task is executed in a one-time unit; the vertical axis shows the CPUs that were simulated; and the timeline shows how tasks were assigned to CPUs, using different colours for each annotated task.

Pregunta 2

Completa

Puntuació 1,00 sobre 1,00

Marca la pregunta

The results reported in Table 3.1 for 3DFFT reveal that: (select the one that is true):

- V1 does not provide any additional parallelism because tasks are still too large to obtain benefit from their parallel execution.
- V2 provides some parallelism, but it is limited only by the value of constant N. Increasing the value for N to 100 (10 times larger than defined in the program) would also increase the parallelism that is obtained with respect to V1 by a factor of 10.
- V3 is exploiting all the available parallelism inside computational tasks (the ones performing FFT and transpositions). Increasing the value for N to 100 (10 times) would not result in any additional benefit in terms of parallelism since task "init_complex_grid" is not further decomposed in finer tasks.
- V4 demonstrates one of the limitations of Tareador: it is not able to simulate task decompositions that could make use of more than 10 processors.
- All the statements above are false.

Pregunta 3

Completa

Puntuació 1,00 sobre 1,00

Marca la pregunta

If we compare V4 and V5 we observe that: (select those that are true):

- [ ] V5 and V4 have the same speedup for 128 processors since they have the same dependencies between init, fftw and transpose tasks, and therefore, the critical path is the same.
- [ ] V5 is more than 4 times faster than V4 for 128 processors because V5 parallel strategy removes some dependencies of the TDG of V4.
- [x] V5 parallel strategy is faster than V4 for a large number of processors because V5 parallel strategy creates tasks with finer granularity than V4 parallel strategy.
- [x] Assume that all tasks are sequentially created by a thread master. In this case, and depending on the relative task creation cost compared to the granularity of the tasks, a real parallel implementation of V4 parallel strategy may be faster than V5 parallel strategy due to the huge number of tasks created.
- [ ] All the statements above are false.